

Platelet Concentration Is Not Associated With Improved Pain for Partial-Thickness Rotator Cuff Tears Treated With Platelet-Rich Plasma

A Systematic Review and Meta-analysis With Meta-regression of Level 1 Randomized Controlled Trials

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Background: Platelet-rich plasma (PRP) has emerged as a nonoperative treatment option to manage pain and improve functionality in patients with partial-thickness rotator cuff tears. Previous studies examining the efficacy of PRP for the management of partial-thickness rotator cuff tears describe heterogeneous results.

Purpose: To determine whether platelet concentration of a PRP injection as management for partial-thickness rotator cuff tears affects visual analog scale (VAS) pain scores.

Study Design: Systematic review; Level of evidence, 1.

Methods: A systematic review was performed by searching PubMed, the Cochrane Library, and Embase to identify level 1 randomized controlled trials (RCTs) that evaluated the clinical efficacy of PRP for the management of partial-thickness rotator cuff tears. The search phrase used was “rotator cuff partial platelet rich plasma.” Included RCTs were classified as utilizing high-dose or low-dose PRP based on their platelet concentration factor as greater than a 4-fold increase over whole blood. The primary outcome evaluated was the standardized mean differences of VAS scores. Mixed-effects and random effects meta-analyses were performed along with meta-regression to evaluate relationships in VAS scores associated with PRP platelet dose.

Results: Ten studies met inclusion and exclusion criteria, including a total of 635 patients (PRP, $n = 307$; control, $n = 328$). The mean patient age at the time of injection was 50.6 years, the mean follow-up time was 10.0 months, and the mean male sex distribution was 44.4%. Of the 10 studies, 6 studies utilized high-dose PRP and 4 studies utilized low-dose PRP. In both the meta-analysis and the meta-regression analyses, there were no significant differences in mean postinjection VAS scores between platelet dosage groups ($P = .36$ and $P = .37$, respectively).

Conclusion: Among level 1 RCTs reporting VAS pain scores after PRP injection for partial-thickness rotator cuff tears, platelet concentration was not significantly associated with improved pain outcomes. However, substantial heterogeneity and inconsistent reporting of PRP composition limit the certainty of this finding.

Keywords: rotator cuff; dose; platelet-rich plasma; platelet concentration; partial-thickness tear

Rotator cuff tears are one of the most common debilitating injuries with, particularly, partial-thickness rotator cuff tears accounting for a substantial portion of cases with a 32% prevalence rate within the general population and substantially decreasing one's quality of life.^{2,26} Surgical repair has predominated the management of partial-thickness rotator cuff tears. Though the decision for surgery often depends on symptoms, failed nonoperative treatment, tear depth, and other patient factors, the standard method of care for a tear with $\geq 50\%$ thickness is surgical repair while those of $<50\%$ thickness are managed with debridement.¹³ Beyond the context of surgical management, platelet-rich plasma (PRP) has grown as a popular nonoperative treatment option. Rich in growth factors, PRP therapy plays a key role in promoting tissue repair and regeneration, stimulating cell growth and proliferation via epidermal growth factor and angiogenesis via vascular endothelial growth factor, as well as increased collagen synthesis and remodeling.²⁶

Though in its translation to the clinic, previous studies have demonstrated heterogeneity in the postinjection clinical findings as treatment for partial-thickness rotator cuff tears.^{20,31} These variable findings of the clinical efficacy of PRP have been consistent in several orthopaedic pathologies. As a result, studies have begun assessing components of PRP composition to explain this variability from which several have found an association of platelet concentration with clinical efficacy in knee osteoarthritis^{4,5} and lateral epicondylitis.²⁴ Regarding PRP for partial-thickness rotator cuff tears, the influence of platelet concentration of PRP on pain is yet to be assessed. Therefore, the purpose of this study is to determine whether platelet concentration of a PRP injection as management for partial-thickness rotator cuff tears affects improvement in pain. We hypothesized that higher platelet concentrations of PRP injections will be associated with improved pain.

METHODS

This systematic review was conducted according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines using a PRISMA checklist. This review was not registered, and access to the review protocol was not prepared. Two independent reviewers (J.J.L. and J.W.B.) searched PubMed, Embase,

and the Cochrane Library up to March 24, 2026. The electronic search strategy used was "rotator cuff partial platelet rich plasma." A total of 640 studies were reviewed by title and/or abstract to determine study eligibility based on inclusion criteria. In cases of disagreement, a third reviewer (R.M.F.) made the final decision. Inclusion and exclusion criteria followed the PICOS (participants, interventions, comparators, outcomes, study design) strategy. Studies selected for inclusion met the following criteria: (1) participants: patients with partial-thickness rotator cuff tears; (2) intervention: PRP injection; (3) comparator: alternative injection or physical therapy; (4) outcomes: clinical efficacy and adverse events; (5) study design: level 1 randomized controlled trials (RCTs) that were published in English. Exclusion criteria included level 2 to 5 studies that did not meet the aforementioned inclusion criteria and studies that did not report adequate detail on PRP composition. Ten studies were determined to meet inclusion criteria (Figure 1). Data extraction from each study was performed independently (E.A.S. and J.W.B.) and then reviewed by a second author (J.J.L.). During the cases where the final platelet concentration of the administered PRP or platelet concentration increase factor was not reported,^{27,29,30,31} the platelet concentration increase factor was calculated as the initial volume of blood collected for PRP preparation divided by the volume of PRP obtained.²⁴ The primary outcome assessed was the visual analog scale (VAS) pain score, which was the most consistently reported outcome with 8 of the 10 studies reporting this specific outcome.^{6,14,17,20,27,29,30,31} There was no need for funding or a third party to obtain any of the collected data. Risk of bias was assessed according to the Cochrane risk-of-bias tool,¹⁵ which incorporates an assessment of randomization, blinding, completeness of outcome data, selection of outcomes reported, and other sources of bias.

Study Methodology Assessment

The Modified Coleman Methodology Score (MCMS)⁹ was used to evaluate study methodology quality. The MCMS has a scaled potential score ranging from 0 to 100. Scores ranging from 85 to 100 are excellent, 70 to 84 are good, 55 to 69 are fair, and <55 are poor. The primary outcomes assessed by the MCMS are study size and type, follow-up time, attrition rate, number of interventions per group, and proper description of study methodology.

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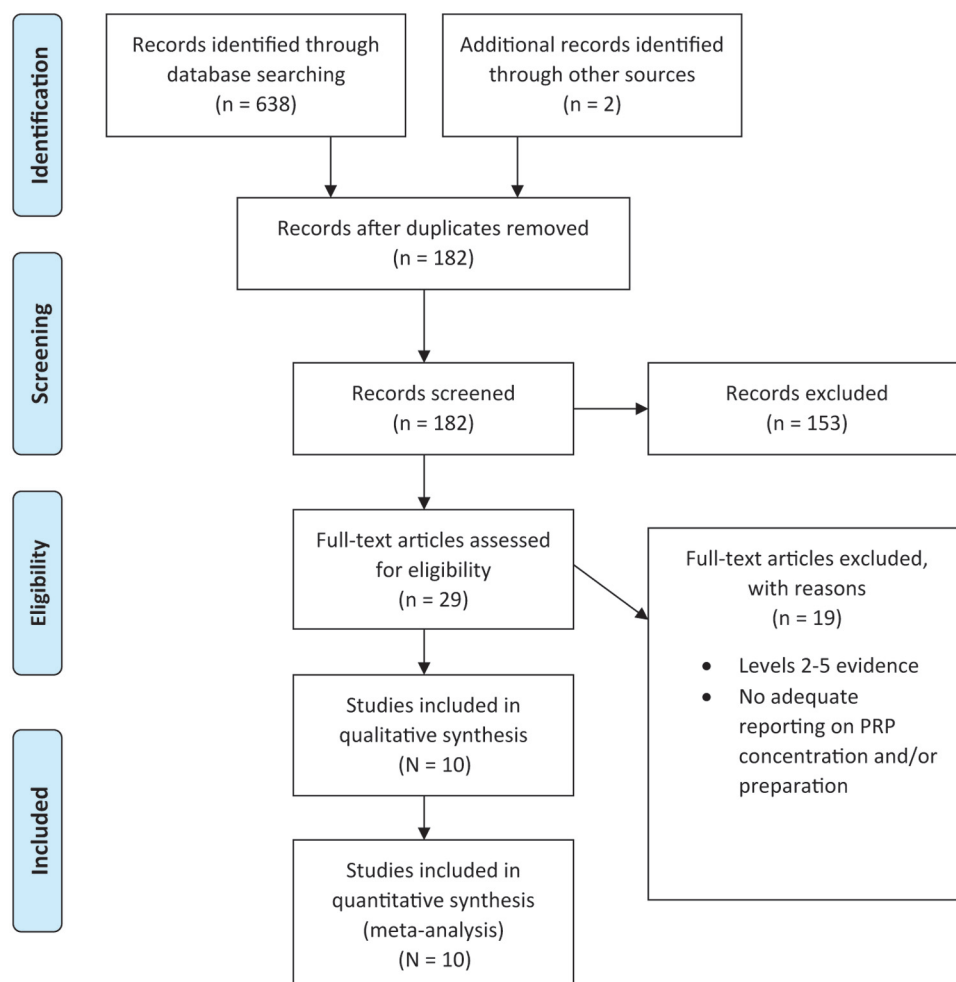


Figure 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram. PRP, platelet-rich plasma.

High-Dose PRP and Low-Dose PRP Terminology

High-dose PRP and low-dose PRP follow similar definitions and cutoff points from Oeding et al,²⁴ a previous meta-analysis evaluating the effect of platelet concentration on the clinical efficacy of PRP for patients with lateral epicondylitis. A platelet concentration increase factor of >4 over whole blood was deemed as high-dose PRP whereas PRP with less than the 4-fold increase was deemed as low-dose PRP. The cutoff point was determined with a previous knowledge that whole-blood platelet concentration is roughly $150 \times 10^3/\mu\text{L}$ to $450 \times 10^3/\mu\text{L}$ as well as the current understanding of the literature defining PRP as having a platelet concentration 3 to 5 times that of whole-blood platelet concentration.^{11,12,22} Therefore, applying what is understood from the literature, we decided it was reasonable to follow the definitions established by Oeding et al²⁴ regarding their determination of high-dose PRP and low-dose PRP.

Statistical Analysis

Mixed-effects meta-analysis models were constructed using Hedges g (bias-corrected standardized mean difference [SMD], calculated with exact formulae), where the difference was based on the postoperative values between the 2 dosage groups. Pooled estimates were derived with the inverse variance method, and between-study variance (τ^2) was estimated using restricted maximum likelihood. Confidence intervals for τ^2 and τ were obtained via the Q-profile method, and heterogeneity was quantified using I^2 statistics derived from Cochran Q . To enable adequate control of heterogeneity, the model determined as the primary analysis in the meta-analysis was the random-effects model while a common-effects model was run as a secondary analysis. In the case where standard deviations were not provided,^{14,27} a quarter of the mean was used as the standard deviation, as previously described.³²

TABLE 1
Included Studies^a

Study	LOE	N	Patient Age	PRP Type	Platelet Concentration Increase Factor	Platelet Concentration ($\times 10^3/\mu\text{L}$)	Minimum Follow-up, mo	Sex Distribution, % Male
Akan et al, 2019 ¹	1	30	50.4 $\pm$ 9.4	NR	5.0	NR	12	63.3
Cai et al, 2019 ⁶	1	45	40.6 $\pm$ 7.9	LP-PRP	3.3	1000	12	48.9
Guity et al, 2025 ¹⁴	1	39	60.0 $\pm$ 15.0	NR	5.5	NR	6	46.2
Ilhanli et al, 2015 ¹⁷	1	30	59.2 $\pm$ 10.8	LP-PRP	2.3	NR	12	30.0
Kesikburun et al, 2013 ¹⁹	1	20	45.5 $\pm$ 11.8	NR	4.0	1014.9	12	35.0
Kwong et al, 2021 ²⁰	1	47	49.9 $\pm$ 9.7	LP-PRP	1.6	NR	12	34.0
Rha et al, 2013 ²⁷	1	20	52.2 $\pm$ 9.5	NR	8.3	NR	6	45.0
Schwitzgubel et al, 2019 ²⁹	1	41	48.2 $\pm$ 16.2	LP-PRP	4.0	NR	12	53.6
Serya et al, 2021 ³⁰	1	20	52.0 $\pm$ 9.9	NR	8.0	NR	3	55.0
Thepsoparn et al, 2021 ³¹	1	15	51.3 $\pm$ 10.3	LP-PRP	3.0	NR	6	20.0

^aData are presented as mean $\pm$ SD unless otherwise indicated. LOE, level of evidence; LP, leukocyte-poor; NR, not reported; PRP, platelet-rich plasma.

Subgroup analyses were performed by stratifying studies according to PRP platelet dose. To further investigate heterogeneity, meta-regression models were constructed with PRP dose as the moderator. Forest plots were generated to display pooled effect estimates and study-level contributions. Regression coefficients from meta-regression models were reported with standard errors, test statistics, *P* values, and 95% CIs. All statistical analyses were performed using R (Version 4.6.1).

RESULTS

Ten studies met inclusion and exclusion criteria (Figure 1), including a total of 635 patients (PRP, *n* = 307; control, *n* = 328). The mean patient age at the time of injection was 50.6 years, the mean follow-up time was 10.0 months, and the mean male sex distribution was 44.4%. Of the 10 studies, 6 studies utilized high-dose PRP and 4 studies utilized low-dose PRP (Table 1).

PRP Preparation and Administration Method

All 10 included studies described details of PRP preparation. Specific machines or processing kits used for PRP preparation were highly variable but were all reported as a centrifugation system. One study⁶ reported using the Labofuge 400R Heraeus (Thermo Fisher Scientific). One study¹⁹ reported using the GPS III Platelet Separation

System (Biomet Biologics). One study²⁷ reported using the Prosys PRP Platelet Concentration System (Tozaiholdings Inc). One study³⁰ reported using GD medical pharma PRP kits. One study³¹ reported using the ACP Double Syringe System (Arthrex). Two studies^{20,29} reported using the Regen Kit (Regen Lab). Three studies^{1,14,17} did not report what PRP preparation kit was utilized. All studies reported drawing blood from the peripheral antecubital vein while also adequately describing the spin details of the centrifugation to isolate the red blood cells from the upper plasma layer. In 5 studies^{1,6,14,27,30} (50.0%), the supernatant layer was transferred to a new centrifuge tube for a second centrifugation to separate the platelet-poor plasma layer from the PRP layer and all studies reported these second spin details. In 2 studies^{17,30} (20.0%), the PRP was then activated by adding calcium chloride, while 4 studies^{1,19,20,29} reported no activation. The remaining studies did not report any details on activation.^{6,14,27,31} Among the included studies, platelet concentration was reported between 1.6 and 8.0 times baseline values. Eight of the studies^{1,14,17,20,27,29,30,31} (80.0%) did not report on the final injected platelet concentration.

Regarding the administration method, 9 studies[§] reported using ultrasound guidance for visualization of the injection site. In 4 studies,^{14,19,20,31} PRP was adminis-

[§]References 1, 6, 14, 19, 20, 27, 29, 30, 31.

TABLE 2
Modified Coleman Methodology Score (MCMS)

Study	MCMS Score
Akan et al, 2019 ¹	82
Cai et al, 2019 ⁶	89
Guity et al, 2025 ¹⁴	85
Ilhanli et al, 2015 ¹⁷	91
Kesikburun et al, 2013 ¹⁹	85
Kwong et al, 2021 ²⁰	86
Rha et al, 2013 ²⁷	75
Schwitzguebel et al, 2019 ²⁹	86
Serya et al, 2021 ³⁰	88
Thepsoparn et al, 2021 ³¹	81
Mean	84.8 ± 4.6

tered once at the beginning of the study. Two studies^{17,30} reported administering the PRP a total of 3 times at 1-week consecutive intervals. Two studies^{27,29} reported administering the PRP a total of 2 times at 4-week intervals. Akan et al¹ reported administering PRP a total of 3 times at 3-week intervals. Cai et al⁶ reported administration of PRP for a total of 4 times at 1-week consecutive intervals. Five studies^{1,27,29,30,31} reported administering the injection at the site of the tendon tear. Three studies^{6,14,19} reported administering the injection into the subacromial space, of which 2 of the 3 studies^{14,19} utilized a posterolateral approach. One study²⁰ reported administering the injection at the site of the tendon as well as within the subacromial space. One study¹⁷ reported administering the injection into the intra-articular space.

Modified Coleman Methodology Score

Table 2 shows the MCMS scores from the 10 included studies. Seven studies^{6,14,17,19,20,29,30} received excellent scores and 3 studies^{1,27,31} received a good score.

Methodologic Quality Assessment

The results of the methodologic quality assessment of included studies using the Cochrane risk-of-bias tool are presented in Figure 2. Sequence generation was adequately reported by all studies (low risk of bias). In all studies, concealment of allocation from the investigators was described (low risk of bias). All studies were deemed to be at low risk for detection bias because of the blinding of the outcome assessor. Patients in all studies were blinded to their intervention group (low risk of bias). Two studies^{17,27} reported a minor loss of follow-up between 10% and 20% without proper explanation (unclear risk of bias), while no other studies reported significant loss of follow-up (low risk of bias).

Visual Analog Scale

In the overall meta-analysis for VAS pain scores, there were 8 studies^{6,14,17,20,27,29,30,31} with 550 observations (257 experimental, 293 control). The pooled SMD was -1.09 (95% CI, -2.94 to 0.75) under the primary random-effects model and -0.16 (95% CI, -0.36 to 0.03) under the secondary common-effects model (Table 3). Between-study heterogeneity was substantial ($\tau^2 = 6.9942$, 95% CI $[2.9757$ to $30.2626]$; $I^2 = 97.4\%$, 95% CI $[96.3$ to $98.2]$) (Figure 3), and the test of heterogeneity was statistically significant ($Q = 271.54$; $df = 7$; $P < .0001$), suggesting true differences in effect sizes across studies.

In subgroup analyses, in the high-dose subgroup ($k = 4$), the pooled effect size was 0.20 (95% CI, -0.07 to 0.48), while in the low-dose subgroup ($k = 4$), the pooled effect size was -0.53 (95% CI, -0.81 to -0.25) in the common effect model (Table 4; Figure 3).

Heterogeneity was substantial within both the high-dose ($I^2 = 97.3\%$) and low-dose subgroups ($I^2 = 98.0\%$). In the random-effects model, the high-dose subgroup ($k = 4$) had a pooled effect size of -0.22 (95% CI, -2.02 to 1.58) while the low-dose subgroup ($k = 4$) had a pooled effect size of -1.98 (95% CI, -5.30 to 1.34). Tests for subgroup differences were not statistically significant under the random-effects model ($Q = 0.83$; $df = 1$; $P = .36$) (Figure 3). Given the substantial heterogeneity, the random-effects model findings were prioritized.

In meta-regression analyses, platelet dose was not significantly associated with VAS scores (Table 5).

DISCUSSION

The primary finding of the present study was that platelet concentration of the injected PRP was not associated with improved pain outcomes for patients with partial-thickness rotator cuff tears. This finding is supported by the random-effects model, which was determined to be the primary, more appropriate statistical framework from the analytical output. The common-effects model framework was not prioritized, because the assumption of a shared true effect size is not met and therefore does not accurately fit the high heterogeneity observed by the included studies (97.3% and 98.0% in the high-dose and low-dose groups, respectively). Therefore, in the case where no significance was observed in the random-effects output but under the common-effects model, a lower PRP platelet concentration was significantly associated with less pain than a higher PRP platelet concentration, the random-effects model's nonsignificant findings were determined to be the primary output from the meta-analysis. The observed high heterogeneity is further highlighted by the extensive clinical and methodological variability in PRP preparation, injection protocols, number of injections, PRP composition, as well as several other variables within PRP orthopaedic investigations. It is therefore additionally

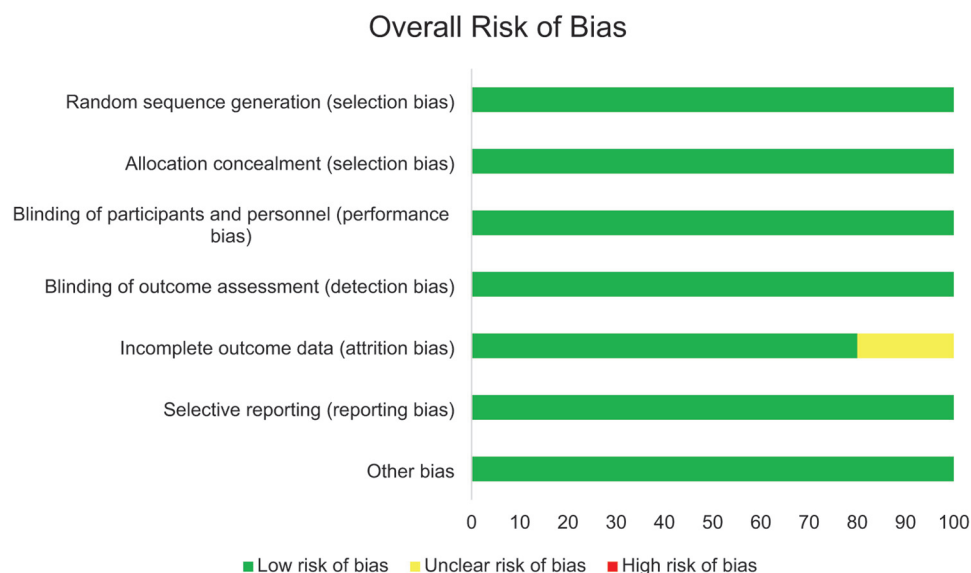


Figure 2. Risk-of-bias graph. Risk of bias is presented as a percentage across all included studies (green, low risk; yellow, unclear; red, high risk).

TABLE 3
Summary of Overall Meta-analyses for VAS^a

Outcome	Model	Studies	Observations	Effect Size (95% CI)	<i>P</i>	τ^2	I^2 , %
VAS	Common	8	550	-0.16 (-0.36 to 0.03)	.10	6.99	97.4
	Random	8	550	-1.09 (-2.94 to 0.75)	.25	6.99	97.4

^aVAS, visual analog scale.

TABLE 4
Summary of Subgroup Meta-analyses for VAS^a

Outcome	Model	Subgroup	Subgroup <i>k</i>	Subgroup Effect (95% CI)	Subgroup I^2	<i>P</i> (subgroup difference test)
VAS	Common	High-Dose	4	0.20 (-0.07 to 0.48)	97.3%	.0002
	Common	Low-Dose	4	-0.53 (-0.81 to -0.26)	98.0%	
	Random	High-Dose	4	-0.22 (-2.02 to 1.58)	$\tau^2 = 3.28$	.36
	Random	Low-Dose	4	-1.98 (-5.30 to 1.34)	$\tau^2 = 11.34$	

^aVAS, visual analog scale.

difficult to justify the common-effects model assumption is met and its associated findings to be appropriate within the current study. As a result, the evaluation of the platelet concentration impact on pain improvement for those with partial-thickness rotator cuff tears were better assessed under the random-effects model. Despite basic science studies observing higher platelet concentration to promote growth factors and greater healing at the injury site,¹⁰ this finding does not support the idea that a PRP platelet concentration 4 times greater than that of whole blood correlates with greater improved pain in the context of rotator cuff tears. Therefore, just as it is not yet

understood whether leukocyte-poor or leukocyte-rich PRP is superior, platelet concentration of PRP has yet to demonstrate influence on outcomes and demonstrate superiority.

In the case of tendinopathy and tears, it was found that in a dose-dependent manner, PRP stimulated cellular proliferation, amplified gene expression, and increased collagen levels within tenocytes to maximize the synthesis of the tendon matrix in degenerative rotator cuff tendon tears.¹⁸ Ultimately, higher PRP doses would presumably promote these anabolic activities at a much greater level than that of lower PRP doses, translating to improved pain in patients with rotator cuff tears. However, given

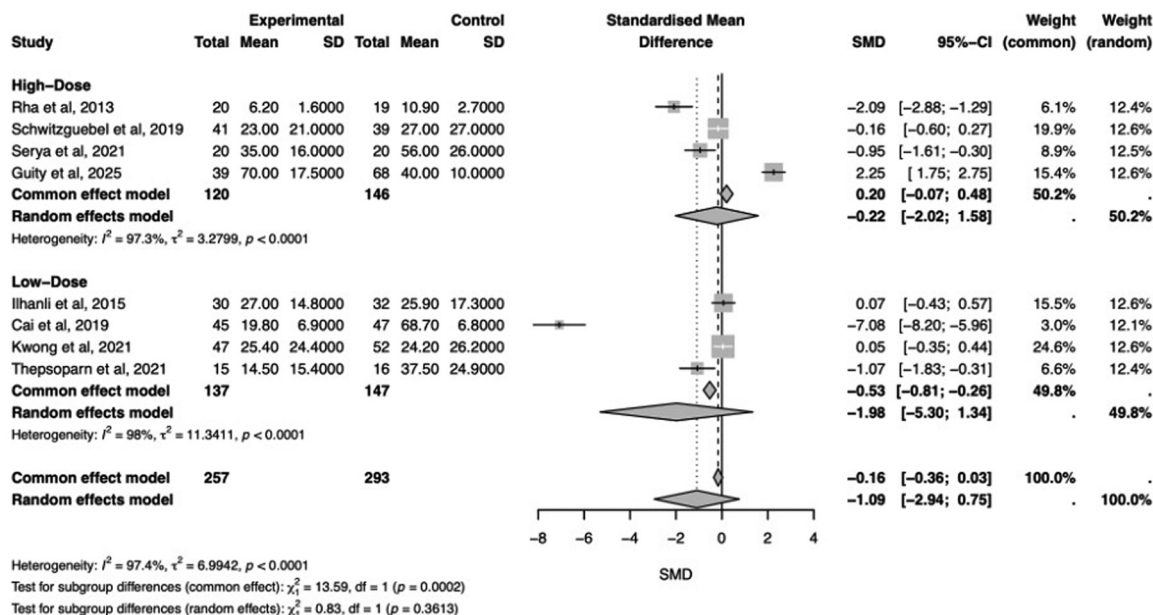


Figure 3. Forest plot of visual analog scale scores. A common-effects model and random-effects model were run by subgroup (high dose and low dose) as well as overall for all included studies. SMD, standardized mean difference.

TABLE 5
Meta-regression Results Evaluating the Effect of Platelet-Rich Plasma Dose (low vs high) on VAS Scores^a

Outcome	Statistic	Coefficient	SE	Statistic	P	95% CI
VAS	SMD	-1.736	1.917	-0.906	.37	(-5.49 to 2.02)

^aCoefficients represent the difference between low-dose and high-dose groups for continuous outcomes. SMD, standardized mean difference; VAS, visual analog scale.

the current study's findings, the translational aspect is yet to be observed, fully explored, and supported. The probable rationale for this may be related to the alteration of tissue characteristics and thereby potential opposing activity of tendon healing. Carr et al⁷ observed PRP to reduce the rotator cuff tendon tissue vascularity and cellularity while increasing apoptosis. All these biological mechanisms of PRP are heavily influenced by the preparation methods that have significantly varied among studies where some may be promoting the anabolic effect while others may foresee the alteration of tissue in an unfavorable manner. Platelet concentrations contribute extensively to this variation, in understanding which concentration may optimize the regenerative healing processes of PRP to then minimize patient recovery time, though its influence remains unclear.

Given PRP is not approved by the Food and Drug Administration and lacks coverage by most insurance carriers, patients will have to make very expensive payments to receive PRP. In the context of partial-thickness rotator cuff repairs, the current study demonstrates PRP may not present as the best option in the condition's

management given the significant high cost and lack of significance when compared with alternative nonsurgical treatments. Future studies with longer follow-up are necessary to fully evaluate the variability and clinical efficacy of PRP for the management of partial-thickness rotator cuff tears. Currently, there is yet to be sufficient reason to utilize PRP within this context and validate the expensive out-of-pocket cost of PRP for the patient. In addition to the evidence demonstrated by the current study, there remains questions that require further study to better understand the impact of other aspects of PRP preparation and composition that may be influencing its efficacy as management for partial-thickness rotator cuff tears. For example, some factors include growth factor concentration, leukocyte concentration, PRP activation, PRP administration, volume of PRP, and preparation systems. Such investigations in the form of RCTs are of interest.

It should also be noted that though it was not the primary focus of the study, the overall meta-analysis demonstrated no significant difference in pain between patients who received a PRP injection and those who didn't. This nonsignificant finding adds to the complexity of the

variable findings surrounding the utilization of PRP for the management of rotator cuff tendinopathy and tears.^{8,21,28} The high concentration of platelets, growth factors, and cytokines of PRP promote angiogenesis, matrix deposition, and cellular proliferation thereby supporting the tendon healing process.^{3,25} Therefore, the contradiction against this basic science study observation of PRP adds greater complexity in understanding what differences are contributing to the efficacy of PRP for partial-thickness rotator cuff tears. Importantly, the current study did not include nor observe any RCTs that have directly compared the preparation of injected PRP with a higher platelet concentration to that of a lower platelet concentration in the treatment of partial-thickness rotator cuff tears. Future studies are suggested to assess this gap within the literature.

The certainty of evidence in the current study was assessed informally using the Grading of Recommendations Assessment, Development and Evaluation framework. As all included studies were level 1 RCTs, the body of evidence began at high certainty but was downgraded across several domains. *Risk of bias* was not a major concern: sequence generation, allocation concealment, blinding of patients, and blinding of outcome assessors were low risk across all studies, and the mean MCMS score was 84.8 (good-to-excellent). *Inconsistency* was serious to very serious: between-study heterogeneity for the primary VAS outcome was extremely high ($I^2 = 97.4\%$ overall; 97.3% and 98.0% in the high- and low-dose subgroups), with a statistically significant Q test ($P < .0001$). *Indirectness* was moderate, driven by variability in comparators, injection protocols, and follow-up duration, and by reliance on the VAS score as a single subjective pain outcome rather than a validated composite pain or functional outcome. *Imprecision* was serious: only 8 studies contributed to the pooled VAS analysis, and the random-effects pooled estimate had a wide interval crossing the null in both directions, spanning clinically important benefit and harm. *Publication bias* could not be formally assessed (eg, via funnel plot or Egger test) given the small number of studies but cannot be excluded. An additional concern is exposure misclassification: 80% of studies did not report the final injected platelet concentration, requiring back-calculation from processing volumes in 4 studies, which introduces uncertainty into the dose classification the meta-regression depends upon. Ultimately, these factors support a certainty-of-evidence rating of low to very low for the conclusion that platelet concentration of PRP for partial-thickness rotator cuff tears is not associated with improved pain outcomes.

Limitations

The limitations of this study should be noted. The included studies differed regarding PRP injection administration methods, follow-up periods, rehabilitation protocols, and tear sizes for included patients. Several RCTs with a follow-up period of <12 months were included, leading to highly variable follow-up periods that have ranged from

3 to 12 months. The current study's lack of protocol registration and limited number of studies for meta-regression must be acknowledged as well. In addition, there was high variability in the composition and preparation of PRP, which similarly prevented a subanalysis on this topic from being performed. Furthermore, the platelet concentration increase factor of 4 studies^{27,29,30,31} was not reported and calculated and may not correspond with the actual platelet concentration of the injected PRP, which was also not reported in 8 of the 10 included studies. This limitation of the current study emphasizes the importance of sufficiently reporting PRP composition details to better understand what form of PRP was injected and serve to promote sufficient clear reporting within studies utilizing PRP. To make an attempt in resolving the heterogeneity of PRP within the literature, future studies evaluating PRP are advised to follow reporting guidelines such as the Minimum Information for Studies Evaluating Biologics in Orthopaedics²³ and the most recent consensus of the Biologics Association.¹⁶ Overall, the heterogeneous nature of the included RCTs in their study design, including patient population, PRP preparation, and insufficient reporting of the injected PRP composition, serves as a confounding factor in the current systematic review and meta-analysis.

CONCLUSION

Among level 1 RCTs reporting VAS pain scores after PRP injection for partial-thickness rotator cuff tears, platelet concentration was not significantly associated with improved pain outcomes. However, substantial heterogeneity and inconsistent reporting of PRP composition limit the certainty of this finding.

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